

BaRcelona Summer School for Demography
Module 2: Demography in R

Population Growth

9 July, 2026

Instructor: Tim Riffe



Universidad
del País Vasco



Euskal Herriko
Unibertsitatea

ikerbasque
Basque Foundation for Science



CED^R
CENTRE D'ESTUDIS
DEMOGRÀFICS

Workshop plan, July 6-10, 2026

- 1: Monday Intro concepts, and R setup
- 2: Tuesday Mortality and fertility
- 3: Wednesday Standardization and decomposition
- 4: Thursday Growth**
- 5: Friday Projection

Growth

- Empirical growth
- Stationarity and Stability
- Structure vs size
- Speedometers
- Model abuse

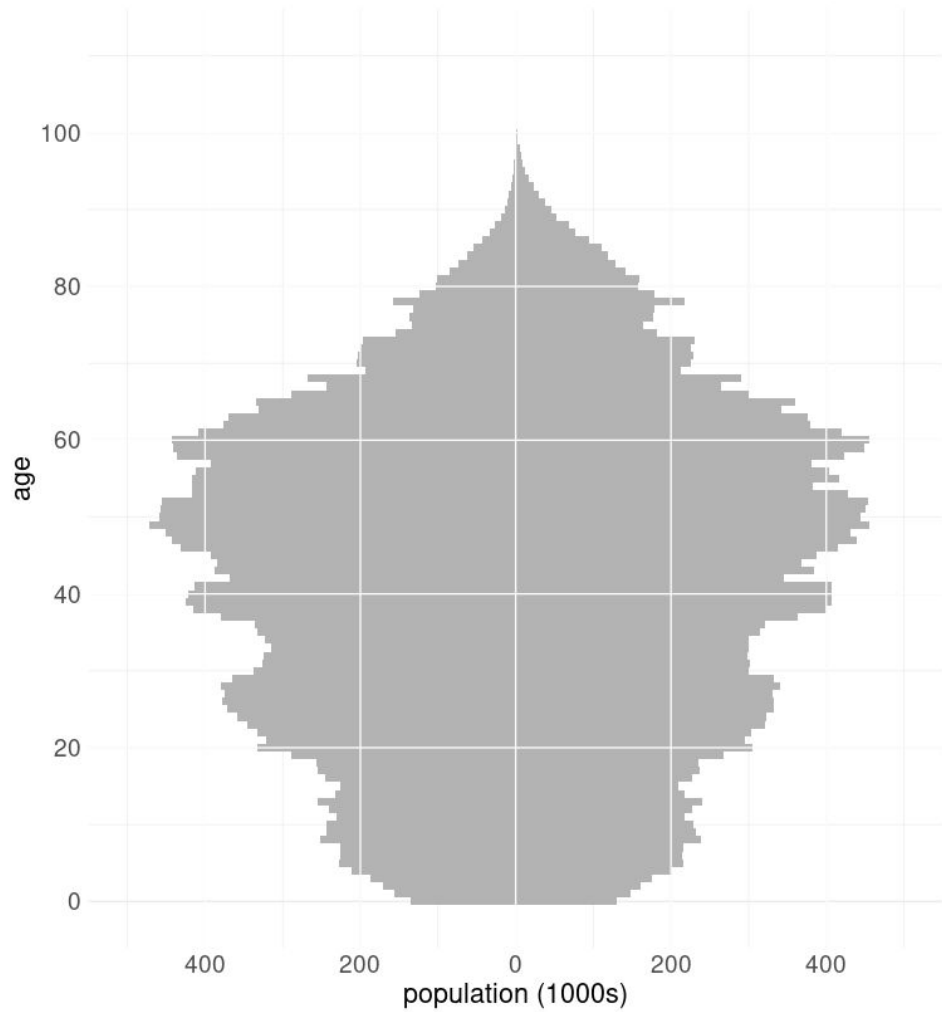
Empirical growth

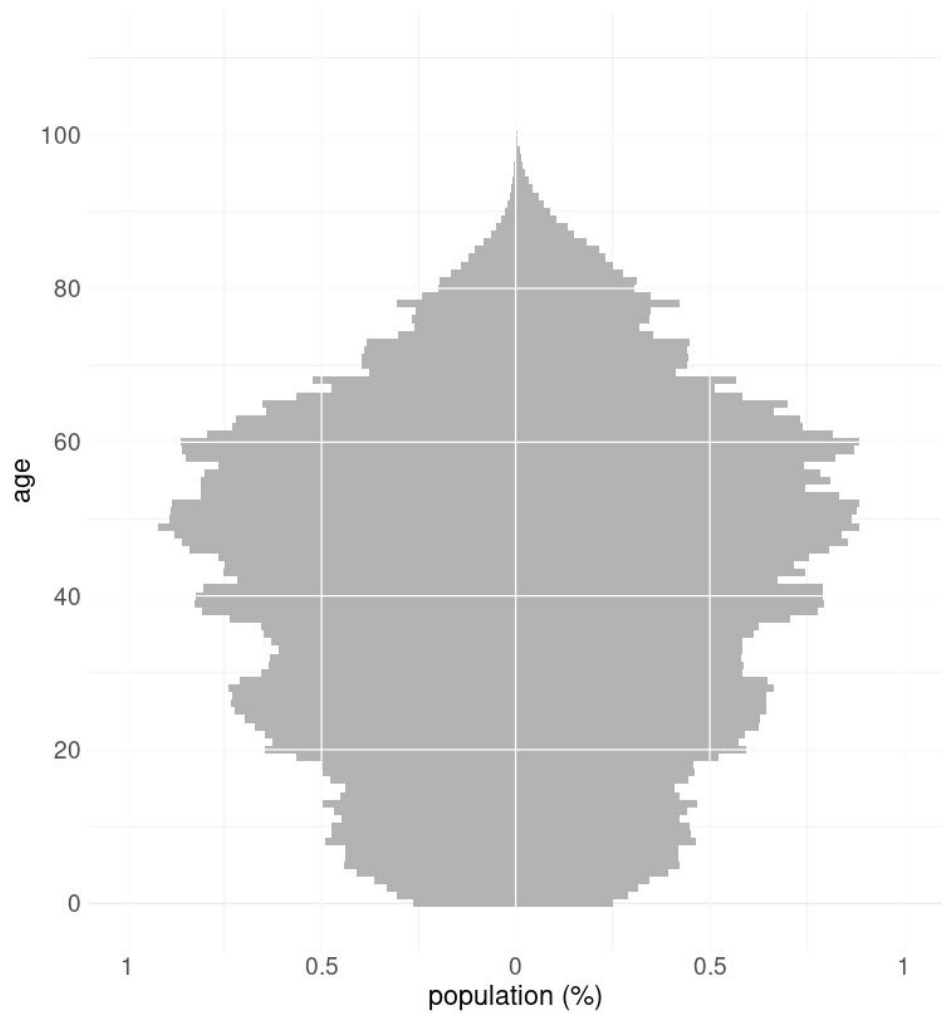
- Geometric

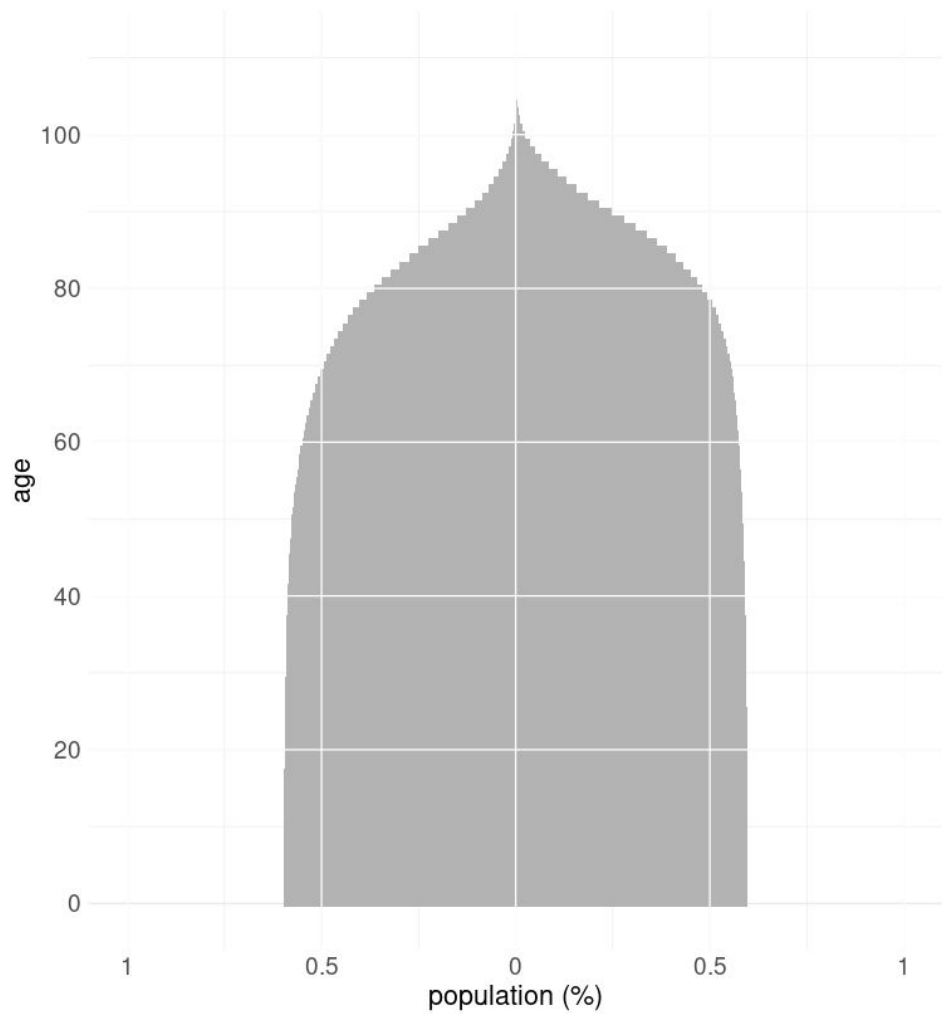
$$r = \frac{N(T) - N(0)}{N(0)} \frac{1}{T}$$

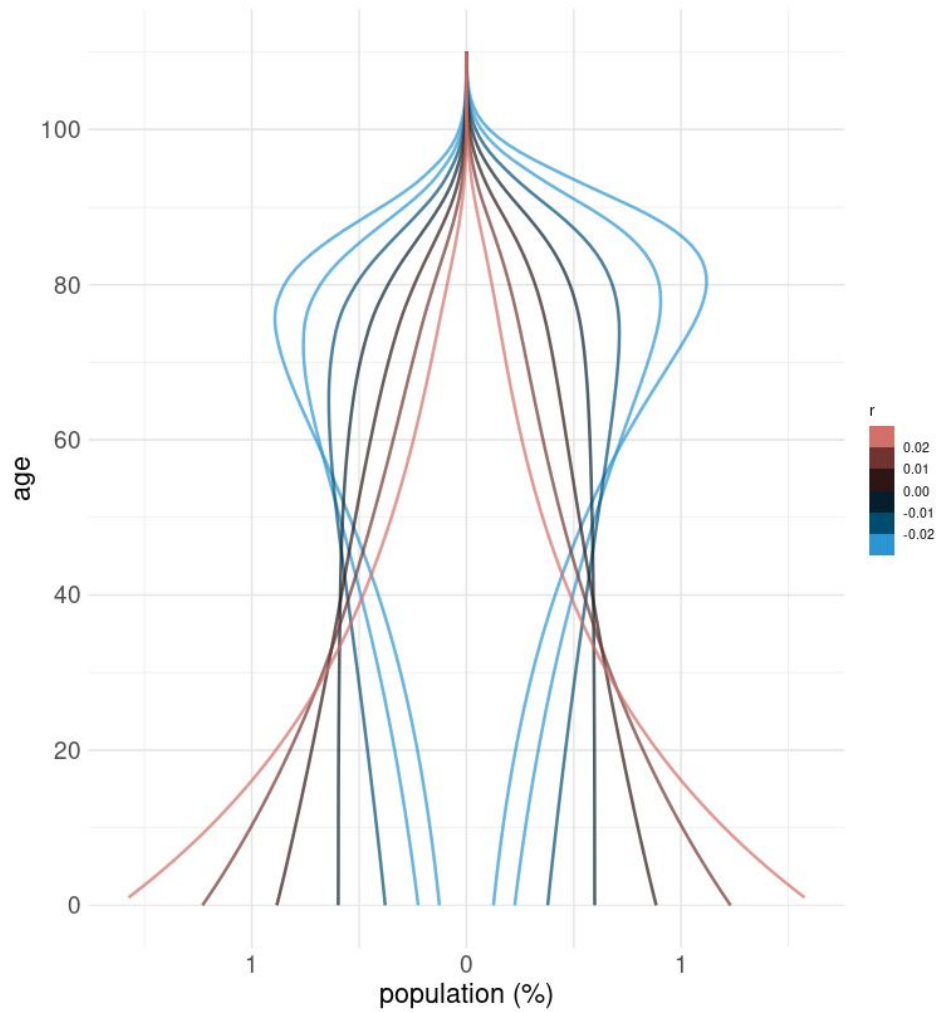
- Exponential

$$r = \frac{\ln\left(\frac{N(T)}{N(0)}\right)}{T}$$









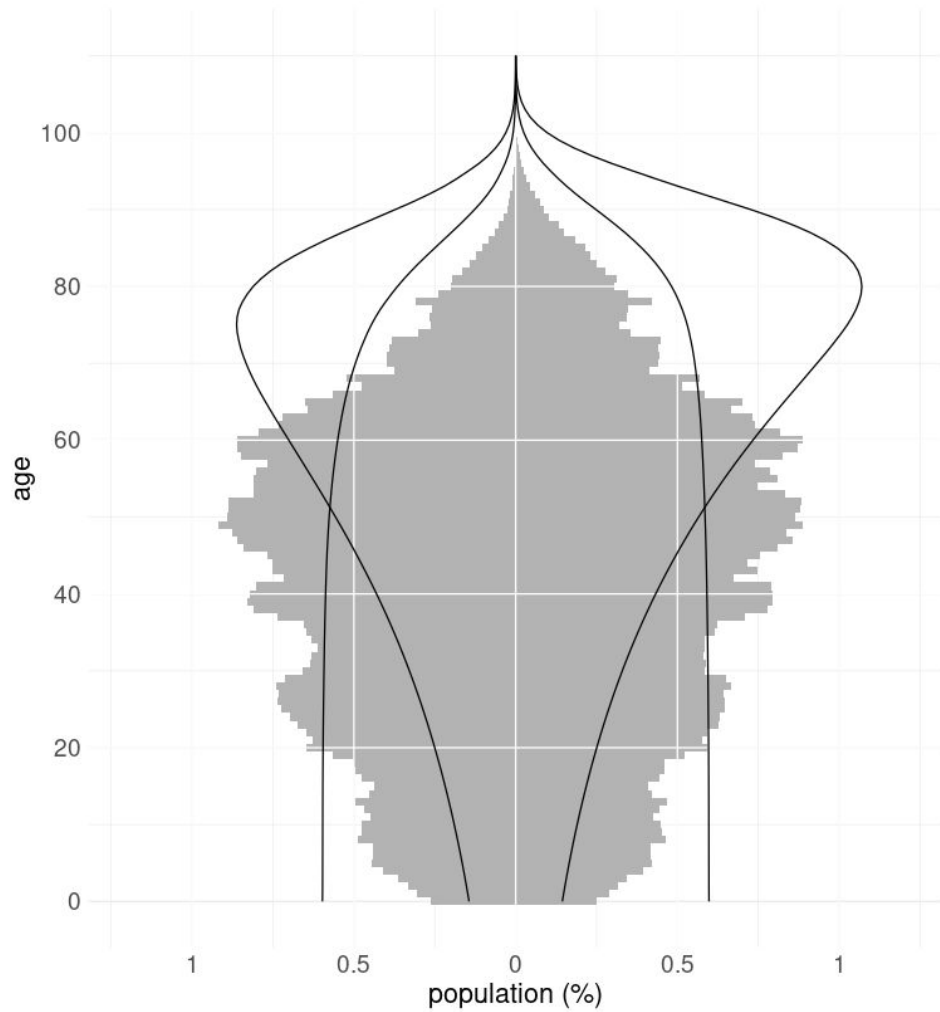




Photo:
Riccardo Pierri

HOME > SCIENCE > VOL. 346, NO. 6206 > IS LOW FERTILITY REALLY A PROBLEM? POPULATION AGING, DEPENDENCY, AND CONSUMPTION



REPORT



Is low fertility really a problem? Population aging, dependency, and consumption

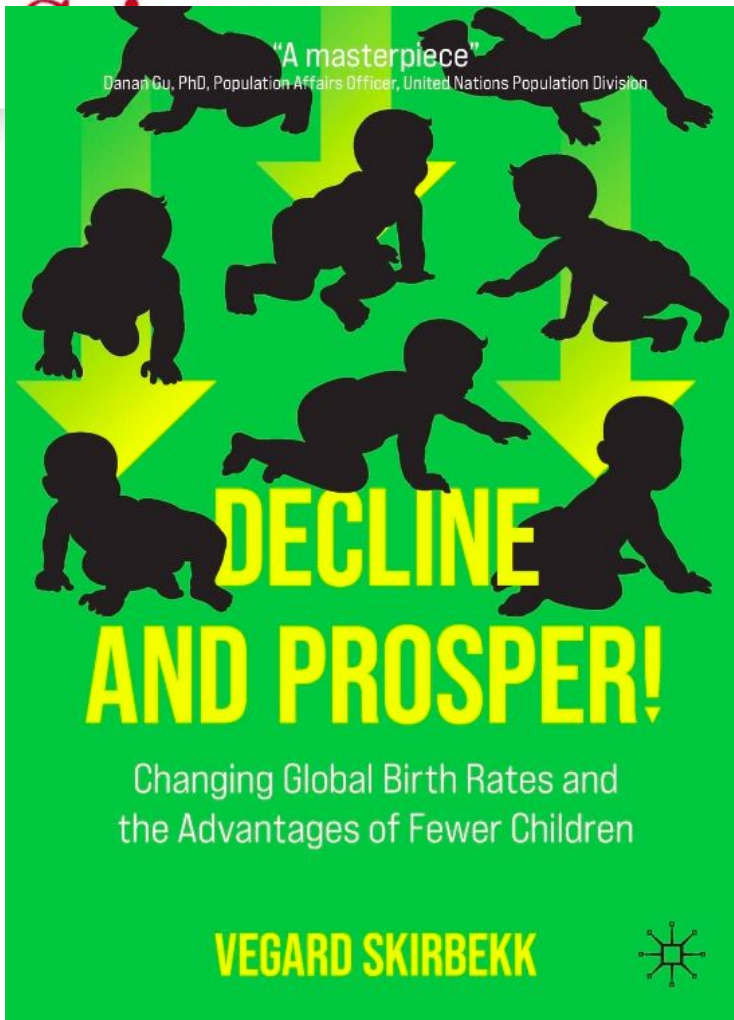
[RONALD LEE](#), [ANDREW MASON](#), [MEMBERS OF THE NTA NETWORK](#), [RONALD LEE](#), [ANDREW MASON](#), [EUGENIA AMPORFU](#), [CHONG-BUM AN](#), [LUIS ROSERO BIXBY](#), [JORGE BRAVO](#), [...], AND [QI ZHANG](#)

+47 authors

[Authors Info & Affiliations](#)

SCIENCE • 10 Oct 2014 • Vol 346, Issue 6206 • pp. 229-234 • DOI: [10.1126/science.1250542](https://doi.org/10.1126/science.1250542)

NTA data: <https://www.ntaccounts.org/web/nta/show/>



Current Issue

First release papers

Archive

About ▾

Submit manuscript

REALLY A PROBLEM? POPULATION AGING, DEPENDENCY, AND CONSUMPTION



problem? Population aging, de- nption

DONALD LEE, ANDREW MASON, EUGENIA AMPORFU, CHONG-BUM AN, LUIS ROSERO BIXBY, JORGE BRAVO

ns

10.1126/science.1250542

NTA data: <https://www.ntaccounts.org/web/nta/show/>